

Cryptic phenotypic variation emerges rapidly during the adaptive evolution of a carbapenemase

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isolate	nucleotide	amino acid	MIC (µg/ml)	line founded
1-1	c814t	H274Y	256	H1
1-2	c814t	H274Y	256	
2-1	c814t	H274Y	256	
2-2	c814t	H274Y	256	
3-1	g532t	D179Y	512	D1
3-2	g532t	D179Y	512	
4-1	g532t	D179Y	512	
4-2	g532t	D179Y	512	
5-1	g532t	D179Y	512	
5-2	g532t	D179Y	512	
6-1	c73a, <u>a105g</u> , <u>c204t</u> , g532t	L28M,D179Y	512	D3
6-2	c73a, <u>a105g</u> , <u>c204t</u> , g532t	L28M,D179Y	512	
7-1	c275t,g532t	T93I,D179Y	256	D4
7-2	t503c	L169P	256	
8-1	t58c,c128t,a533c	F23L,S43F,D179A	512	D6
8-2	t58c,c128t,a533c	F23L,S43F,D179A	512	
9-1	g488a,g532a	R163H,D179N	256	D7
9-2	g488a,g532a	R163H,D179N	256	
10-1	c278t,t508c	P94L,S171P	512	S1
10-2	c278t,t508c	P94L,S171P	512	
11-1	t503c	L169P	256	L1
11-2	c587a,c814t	S196Y,H274Y	512	H5

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9 **Supplementary Table 1. Mutations identified in the initial round of directed**
10 **evolution.** This table lists all nucleotide and amino acid substitutions found in 22
11 isolates obtained from 11 independent replicate experiments. Synonymous mutations
12 are indicated by underlined, italicized nucleotide changes. Isolates used to found the
13 lineages for subsequent rounds of evolution are indicated in the last column.

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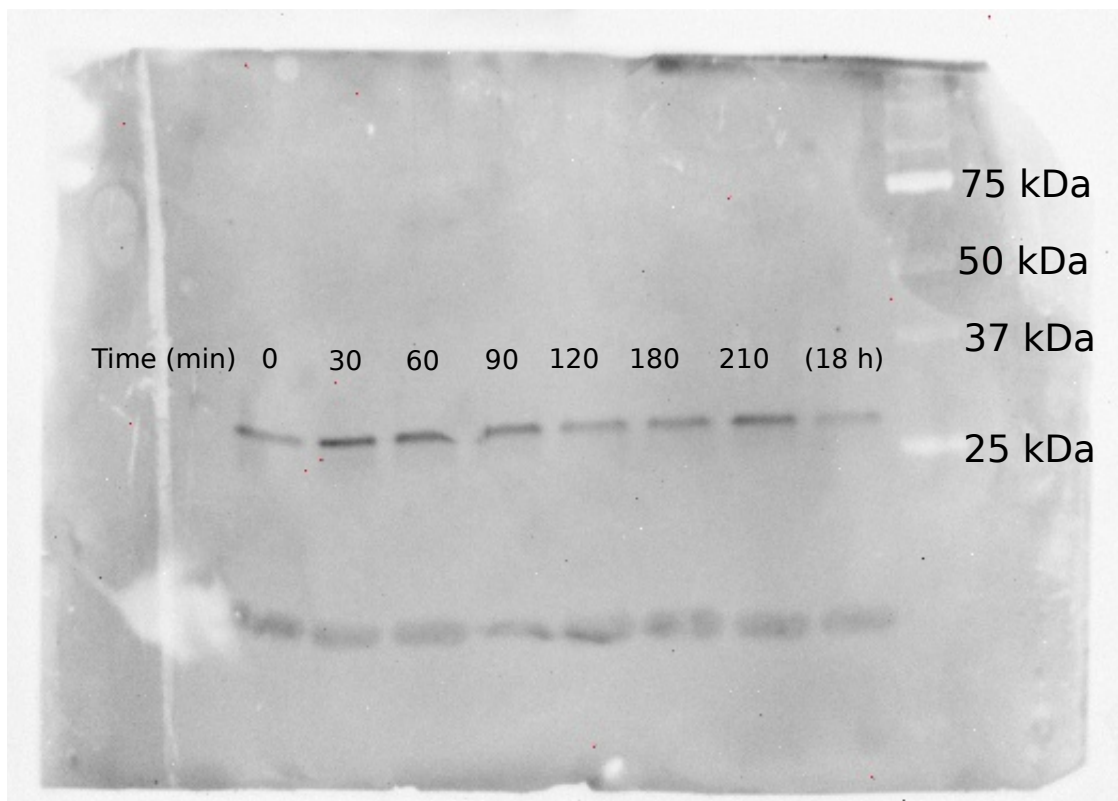
line	Round 1			Round 2			Round 3			Round 4		
	nucleotide	amino acid	MIC	nucleotide	amino acid	MIC	nucleotide	amino acid	MIC	nucleotide	amino acid	MIC
D1	g532t	D179Y	512	<u>c735t</u> , t775a	L260M	1024	c512a, t618a, <u>c735t</u>	A172D, F207L	1024			
D2	g532t	D179Y	512	<u>c735t</u> , t775a	L260M	1024	g436a	G147R	1024			
D3	c73a, <u>a105g</u> , <u>c204t</u> , g532t	L28M, D179Y	512	c278t, <u>g666a</u>	P94L	1024	<u>g346t</u>	none	1024			
D4	c275t, g532t	T93I, D179Y	256	c600t, <u>c654t</u>	A201V	128	<u>c361t</u> , c527t	A177V	128	<u>g267a</u>	none	128
D5	c275t, g532t	T93I, D179Y	256	c725t	T243M	256	<u>t771A</u>	none	256			
D6	t58c, c128t, a533c	F23L, S43F, D179A	512	<u>c278t</u> , c279t, <u>g300a</u>	P94L	1024	c117g, c199a, <u>t775c</u>	D39E, L68M	1024	<u>c507t</u>	none	1024
D7	g488a, g532a	R164H, D179N	256	a500g	E168G	1024	c278a	P94H	1024	c71t	A27V	1024
H1	c814t	H274Y	256	t716c	V240A	2048	g304c	V103L	2048			
H2	c814t	H274Y	256	t716c	V240A	2048	t236a	V80E	4096	none	none	4096
H3	c814t	H274Y	256	c496a, <u>t843c</u>	L167M	256	g455a	R153H	256			
H4	c814t	H274Y	256	c32a	S14Y	256						
H5	c587a, c814t	S196Y, H274Y	512	t716c	V240A	1024	t331c, <u>t621c</u>	Y112H	1024	<u>c195t</u>	none	1024
L1	t503c	L169P	256	c73a, c227t, <u>c663t</u> , c853t	L28M, A77V, L287F	256	a500g, <u>g666a</u>	E168G	256	<u>t738c</u>	none	256
L2	t503c	L169P	256	c278t	P94L	512	g501t	E168D	512			
L3	t503c	L169P	256	c587a, c814t	S197Y, H274Y	512	none	none	512			
S1	c278t, t508c	P94L, S171P	512	<u>a42g</u> , t716c	V240A	1024	g147a, <u>t768a</u>	M49I	2048	c94t	P32S	1024
S2	c278t, t508c	P94L, S171P	512	<u>a42g</u> , t716c	V240A	1024	g147a, <u>t768a</u>	M49I	2048	g22a, <u>c496t</u>	V11I	2048
S3	c278t, t508c	P94L, S171P	512	g822t, <u>a881g</u>	E276D	1024	<u>g675c</u> , <u>g879a</u>	none	1024			

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21 **Supplementary Table 2. Mutations recovered across different rounds of directed**
 22 **evolution.** Lineages are grouped by the presumed first beneficial mutation, color-coded
 23 as blue (D179Y), green (H274Y), yellow (L169P), and red (S171P). Each round
 24 includes three columns showing changes at the nucleotide level in the *bla_{KPC}* gene, at the
 25 amino acid level in the KPC protein, and MIC values for CAZ. Synonymous mutations
 26 are indicated by underlined, italicized nucleotide changes. Note that all mutations
 27 observed in a given round are retained in subsequent rounds, and thus are omitted for
 28 clarity. Details of duplicated lines are shown in gray up to the round in which they were
 29 actually founded. As described in the text, we initially founded nine independent lines in
 30 round 1, then added six duplicates in round 2 (D5, H3, H4, L2, L3, S3), two in round 3
 31 (D2, H2), and one in round 4 (CS2), resulting in a total of 18 evolutionary lines.

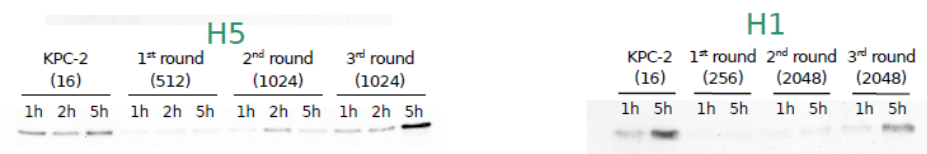
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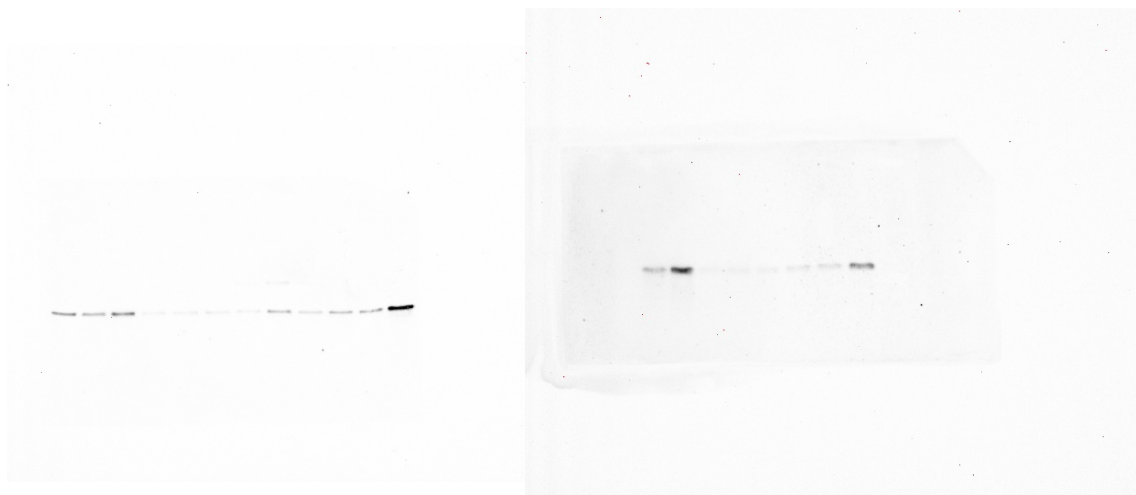


41 **Supplementary Figure 1.** Uncropped and unprocessed gel showing the pilot test with
42 the wild-type KPC-2 enzyme confirming the expected molecular weight of ~29 kDa. We
43 later used the wild-type as a reference marker in downstream assays.

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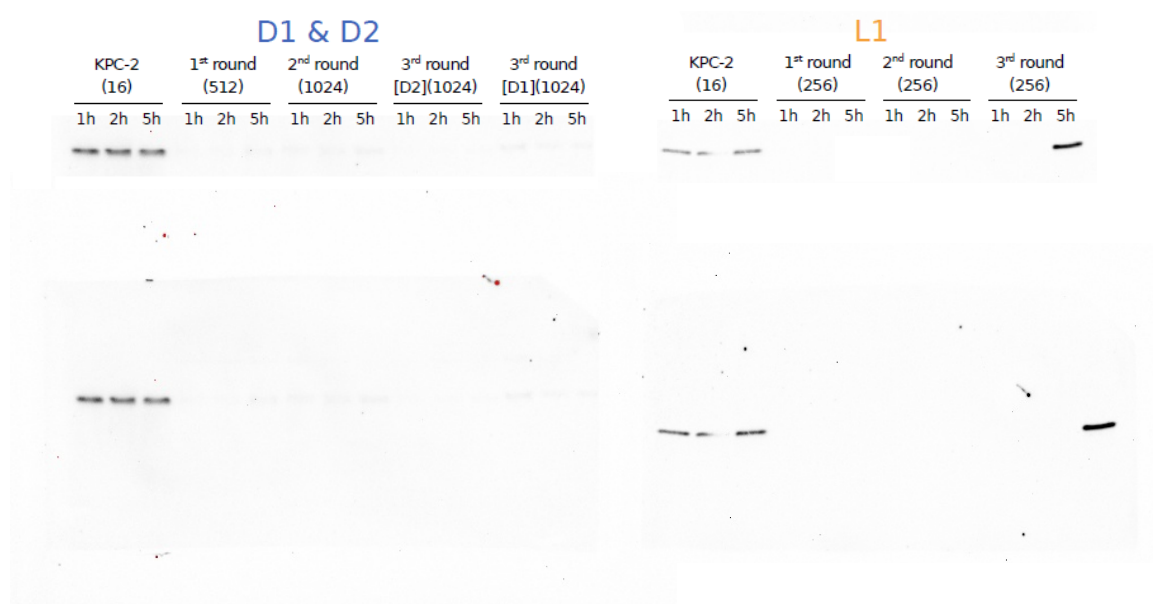


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53 **Supplementary Figure 2.** Uncropped and unprocessed gel for H5 and H1 lines
54 (corresponding section of Ext_Data_FigureS4.pdf on top).

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64 **Supplementary Figure 3.** Uncropped and unprocessed gel for D1, D2 and L1 lines
65 (corresponding section of Ext_Data_FigureS4.pdf on top).

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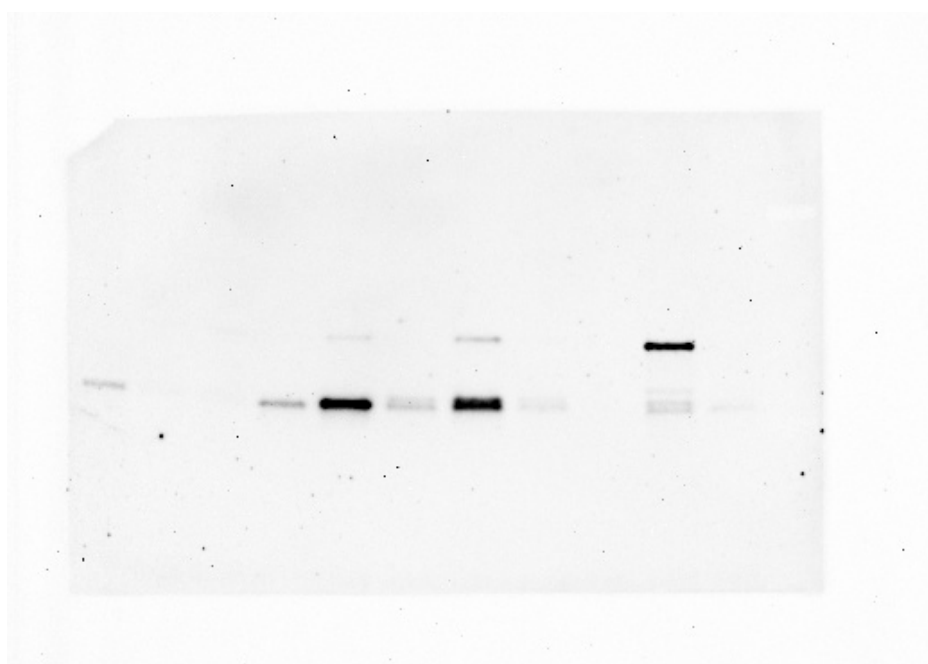


76 **Supplementary Figure 4.** Uncropped and unprocessed gel for S1, S2 and L2 lines
77 (corresponding section of Ext_Data_FigureS4.pdf on top).

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84 **Supplementary Figure 5.** Uncropped and unprocessed gel for the lines assayed at 10 h
85 (corresponding section of Ext_Data_FigureS4.pdf on top).